Then,

$$\Delta_n m = \frac{2}{n}, \tag{C7}$$

$$\Delta_n m^2 = \frac{4m}{n} - \frac{4}{n^2}, \tag{C8}$$

$$\Delta_n m^k = O(\frac{m^{k-1}}{n}). \tag{C9}$$

Now we make the assume the following quadratic ansatz for $\Phi(m)$, for some $a > 0$ and $m_0 \in [-1, 1]$,

$$\Phi(m) = \frac{-a(m - m_0)^2}{2}$$

We have to determine a and m_0 such that the single variable conditionals match as much as possible.

$$\Delta_n \Phi(m) = \frac{-2a(m - m_0)}{n} + \frac{2a}{n^2}. \tag{C10}$$

Now for Ψ, we expand the free energy up to second order around the (as of yet unknown) point m_0,

$$\Psi(m) = -\frac{J}{2}m^2 + hm - S(m_0) - (m - m_0)S'(m_0) - \frac{(m - m_0)^2}{2}S''(m_0) + O((m - m_0)^3) \tag{C11}$$

This gives,

$$\Delta_n \Psi(m) \approx -J(\frac{2m}{n} - \frac{2}{n^2}) + h\frac{2}{n} - S'(m_0)\frac{2}{n} - \frac{2S''(m_0)(m - m_0)}{n} + \frac{2S''(m_0)}{n^2}. \tag{C12}$$

Now equating terms of order $\frac{1}{n^2}$ from (C10) and (C12) gives us the following relation for finding a,

$$a = -J - S''(m_0) \tag{C13}$$

Substituting this back in the previous expression and matching terms of $1/n$ gives us the following expression for m_0

$$-Jm_0 + h - S'(m_0) = 0. \tag{C14}$$

Now this above expression is precisely the first-order stationarity condition for the free energy of CW model. And from $a > 0$ we find that $\Phi''(m_0) = J - S''(m_0) > 0$ which tells us that m_0 lies at the minimum of the free-energy.

This calculation establishes the following important observation: *Minima of the CW free energy defines distributions whose single variable conditionals match that of the CW distribution*

Notice also that $\Phi(m)$ defined as in (C) is just the second order Taylor expansion of the free energy Ψ at m_0. Now starting from this observation we can define a class of potentially interesting metastable states by expanding further around the positive minima of Ψ.

$$\Phi^{(K)}(m) = \sum_{k=2}^{K} \Psi^{(k)}(m_0)\frac{(m - m_0)^k}{k!} \tag{C15}$$

$$\nu^{(K)}(\underline{\sigma}) = \frac{e^{-n\Phi^{(K)}(m(\underline{\sigma}))}}{\binom{n}{\frac{n(1+m)}{2}}}. \tag{C16}$$

1. Numerical evaluation of strong metastability violation in Curie-Weiss models

For distributions which are permuation invariant, i.e. where $\nu(\underline{\sigma})$ is only a function of $m(\underline{\sigma})$, the quantity in (3) can be computed efficiently my mapping the problem on to the magnetization space first.

First let us fix the Markov chain as Glauber dynamics defined by the Curie-Weiss model in (25). This gives us,

$$Pr(\underline{\sigma}_{\neg i}|\underline{\sigma}) = \frac{1}{n}\mu(-\sigma_i|\underline{\sigma}_{\setminus i}) \tag{C17}$$

Now using this the violation in the detailed balance condition as defined in (3) can be rewritten as,

$$\frac{1}{2}\sum_{\underline{\sigma},\underline{\sigma}'}|P(\underline{\sigma}'|\underline{\sigma})\nu(\underline{\sigma}) - P(\underline{\sigma}|\underline{\sigma}')\nu(\underline{\sigma}')| = \frac{1}{2n}\sum_{i=1}^{n}\sum_{\underline{\sigma}}|\mu(-\sigma_i|\underline{\sigma}_{\setminus i})\nu(\underline{\sigma}) - \mu(\sigma_i|\underline{\sigma}_{\setminus i})\nu(\underline{\sigma}_{\neg i})| \tag{C18}$$

$$= \frac{1}{2n}\sum_{i=1}^{n}\sum_{\underline{\sigma}}|(1-\mu(\sigma_i|\underline{\sigma}_{\setminus i}))\nu(\underline{\sigma}) - \mu(\sigma_i|\underline{\sigma}_{\setminus i})\nu(\underline{\sigma}_{\neg i})| \tag{C19}$$

$$= \frac{1}{2n}\sum_{i=1}^{n}\sum_{\underline{\sigma}}|\nu(\underline{\sigma}) - \mu(\sigma_i|\underline{\sigma}_{\setminus i})(\nu(\underline{\sigma}_{\neg i}) + \nu(\underline{\sigma}))| \tag{C20}$$

$$= \frac{1}{2n}\sum_{i=1}^{n}\sum_{\underline{\sigma}}\nu(\underline{\sigma}_{\setminus i})|\nu(\sigma_i|\underline{\sigma}_{\setminus i}) - \mu(\sigma_i|\underline{\sigma}_{\setminus i})| \tag{C21}$$

Where $\nu(\underline{\sigma}_{\setminus i}) = \nu(\underline{\sigma}_{\neg i}) + \nu(\underline{\sigma})$ is the marginal distribution of all spins except σ_i

Now since $|\nu(\sigma_i|\underline{\sigma}_{\setminus i}) - \mu(\sigma_i|\underline{\sigma}_{\setminus i})| = |\nu(-\sigma_i|\underline{\sigma}_{\setminus i}) - \mu(-\sigma_i|\underline{\sigma}_{\setminus i})|$. We can deduce that $\sum_{\underline{\sigma}}\nu(\underline{\sigma}_{\setminus i})|\nu(\sigma_i|\underline{\sigma}_{\setminus i}) - \mu(\sigma_i|\underline{\sigma}_{\setminus i})| = \sum_{\underline{\sigma}}\nu(\underline{\sigma})|\nu(\sigma_i|\underline{\sigma}_{\setminus i}) - \mu(\sigma_i|\underline{\sigma}_{\setminus i})|$. This gives us the final expression,

$$\frac{1}{2}\sum_{\underline{\sigma},\underline{\sigma}'}|P(\underline{\sigma}'|\underline{\sigma})\nu(\underline{\sigma}) - P(\underline{\sigma}|\underline{\sigma}')\nu(\underline{\sigma}')| = \frac{1}{2n}\sum_{i=1}^{n}\sum_{\underline{\sigma}}\nu(\underline{\sigma})|\nu(\sigma_i|\underline{\sigma}_{\setminus i}) - \mu(\sigma_i|\underline{\sigma}_{\setminus i})| \tag{C22}$$

For the Curie-Weiss model, we can easily see that the single variable conditionals have the following expression that depends only on σ_i and the magnetization of the state,

$$\mu(\sigma_i|\underline{\sigma}_{\setminus i}) = \frac{1}{2}(1 - \tanh(\sigma_i(Jm(\underline{\sigma}) - h) + \frac{J}{n}) \equiv f(\sigma_i, m(\underline{\sigma})). \tag{C23}$$

Now we assume a general permutation invariant form for ν,

$$\nu(\underline{\sigma}) = \frac{e^{-n(\Phi(m(\underline{\sigma}))+S(m(\underline{\sigma})))}}{Z_\Phi} \equiv \frac{e^{-nH(m(\underline{\sigma}))}}{Z_\Phi} \tag{C24}$$

From this we can get the conditional as,

$$\nu(\sigma_i|\underline{\sigma}_{\setminus i}) = \frac{1}{1+\frac{\nu(\underline{\sigma}_{\neg i})}{\nu(\underline{\sigma})}} = \frac{1}{1+\exp(n(H(m(\underline{\sigma}) - H(m(\underline{\sigma}_{\neg i})))} \equiv g(\sigma_i, m(\underline{\sigma})) \tag{C25}$$

We can use these expressions for conditionals in (C22) and write this completely in the magnetization space. The number of states with a fixed per spin magnetization m is simply given by $\exp(nS(m))$. Now for a given i, a $\frac{1+m}{2}$

fraction of these will have $\sigma_i = 1$. Using these we can write,

$$\frac{1}{2n}\sum_{i=1}\sum_{\underline{\sigma}}\nu(\underline{\sigma})|\nu(\sigma_i|\underline{\sigma}_{\setminus i}) - \mu(\sigma_i|\underline{\sigma}_{\setminus i})| \tag{C26}$$

$$= \frac{1}{2Z_\Phi n}\sum_{i=1}^{n}\sum_{m\in\{-1,-1+\frac{2}{n},\ldots,1\}} e^{-n\Phi(m)}\left(\frac{1+m}{2}|g(1,m)-f(1,m)| + \frac{1-m}{2}|g(-1,m)-f(-1,m)|\right), \tag{C27}$$

$$= \frac{1}{2Z_\Phi}\sum_{m\in\{-1,-1+\frac{2}{n},\ldots,1\}} e^{-n\Phi(m)}|g(1,m)-f(1,m)| \tag{C28}$$

Where in the last line we have use the normalization condition on the conditionals, $g(1,m)+g(-1,m) = f(1,m)+f(-1,m) = 1$. Using this reformulation we can compute the strong metastability of the state in essentially $O(n)$ time. The normalization constant can also be computed efficiently using the formula, $Z_\phi = \sum_\sigma \exp(-nH(m(\underline{\sigma}))) = \sum_m \exp(-n\Phi(m))$.

The details of numerical evaluation of three different families metastable distributions for the CW model is shown in FIG.2. First we look at the quartic ($K = 4$) and quadratic ($K = 2$) approximations to the CW free energy as defined in (C16). We perform the expansion around the positive minimum of the CW free energy ($m_0 > 0$). As this corresponds to the mode of the distribution with suppressed probability when $h > 0$ (refer FIG.3b). For the third case, we truncate the free energy around m_0 with a certain width,

$$\Phi(m) = \begin{cases} \Psi(m), & \text{if } |m-m_0| < \frac{m_0}{4\sqrt{a}} \\ \infty, & \text{otherwise} \end{cases} \tag{C29}$$

This truncation corresponds to the type of metastable states defined in Section II A. We see numerically that these metastable states have an η value that is exponentially small in the system size.

Appendix D: Proof of Theorem 1

Proof. The $\eta-$strong metastability condition gives us,

$$\frac{1}{2}\sum_{\underline{\sigma},\underline{\sigma}'\in\mathcal{S}}|P(\underline{\sigma}'|\underline{\sigma})\nu(\underline{\sigma}) - P(\underline{\sigma}|\underline{\sigma}')\nu(\underline{\sigma}')| \leq \eta \tag{D1}$$

Now let us introduce the notation $\underline{\sigma}_{i\to\alpha}$ to represent $\underline{\sigma}$ with the i-th variable replaced by the alphabet α.

Now if we consider only one variable transitions of the chain we get,

$$\frac{1}{2}\sum_{\underline{\sigma}\in\mathcal{S}}\sum_{u=1}^{n}\sum_{\alpha\in Q,\alpha\neq\sigma_i}|P(\underline{\sigma}_{u\to\alpha}|\underline{\sigma})\nu(\underline{\sigma}) - P(\underline{\sigma}|\underline{\sigma}_{u\to\alpha})\nu(\underline{\sigma}_{u\to\alpha})| \leq \eta \tag{D2}$$

Since P is a reversible Markov chain, the following relations can be derived from the detailed balance conditions,

$$\frac{P(\underline{\sigma}_{u\to\alpha}|\underline{\sigma})}{P(\underline{\sigma}|\underline{\sigma}_{u\to\alpha})} = \frac{\mu(\underline{\sigma}_{u\to\alpha})}{\mu(\underline{\sigma})} = \frac{\mu(\alpha|\underline{\sigma}_{\setminus u})}{\mu(\sigma_u|\underline{\sigma}_{\setminus u})}. \tag{D3}$$

After plugging these into (D2) and a bit of algebra gives us,

$$\frac{1}{2}\sum_{\underline{\sigma}\in\mathcal{S}}\sum_{u=1}^{n}\sum_{\alpha\in Q,\alpha\neq\sigma_i}\frac{P(\underline{\sigma}_{u\to\alpha}|\underline{\sigma})}{\mu(\alpha|\underline{\sigma}_{\setminus u})}\Big|\mu(\alpha|\underline{\sigma}_{\setminus u})\nu(\underline{\sigma}) - \mu(\sigma_u|\underline{\sigma}_{\setminus u})\nu(\underline{\sigma}_{u\to\alpha})\Big| \leq \eta. \tag{D4}$$